level 2

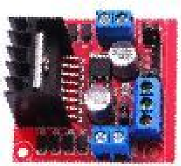
no. of sessions : 5

crane

The crane lifts and moves heavy objects to help people build buildings safely and easily. It is used at construction sites to lift materials like steel beams and concrete blocks.

simple machine: **pulley**
electronic parts: **input vs output**

component gallery



Motor Driver

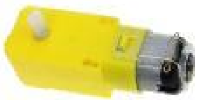
The brain of a robot. I listen to input devices like sensors and tell output devices like motors/LEDs when to turn ON/OFF.

IR Sensor

I am like a robot's eyes. I help robots see objects or follow a path without touching anything.

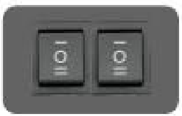
LDR Sensor

I am like a light detector who tells a robot if it is bright or dark and i am used in automatic lights.



Motor

I change electricity into movement and help wheels, fans, and robots spin and move.



DPDT Switch Controller

Like a direction button - i make a motor spin forward or backward without using coding.

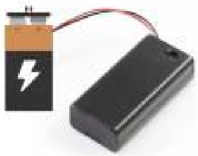
component gallery

Servo Motor

I am a **smart motor** that can stop at the exact position you want. I am used in robot arms, gates, and moving parts.

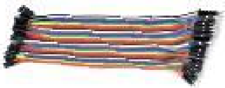
Servo Controller

I am like a **control room** for the servo motor and lets you move the servo easily to test how it works.



Battery & Battery Holder

I **give power** to the project, and my battery holder keeps the battery safe and connected. Together, we help the model work.



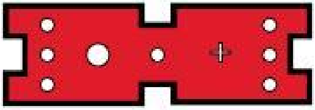
Jumper Wires

We are the **roads for electricity** who connect the different parts of the circuit, so power can travel.

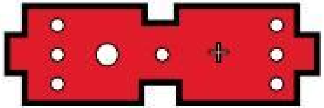
Motor Clamp

I am like a **strong hand** that holds the motor tightly in place while it spins.

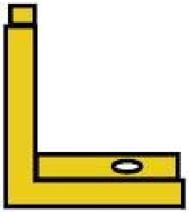
materials needed



F-F beam - x4

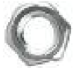


M-M beam - x4



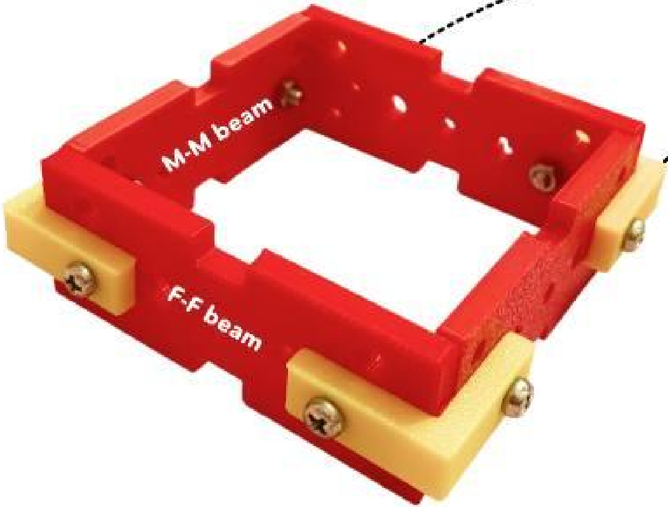
L clamp - x8

small screw - x16



nut - x16

step 1 : assemble the base

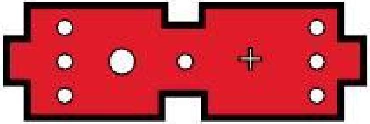


Use **4 Beams** to form a square shape.

Use **L clamps** to hold them together.

🔁 **Repeat step 1 and make another square.**

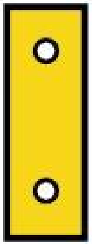
materials needed



M-M beam - x2

small screw - x8

nut - x8



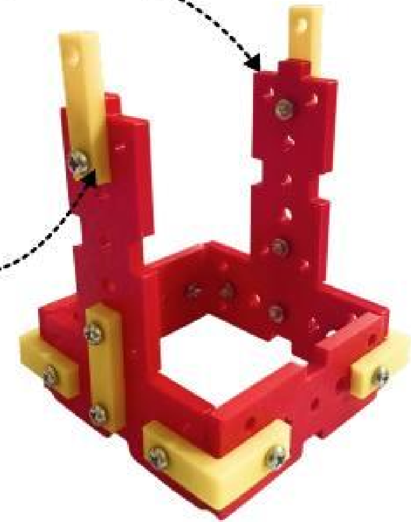
I clamp - x4

step 2: extend the supports for the base

1.

Use 2 **M-M beams** to join the two squares parts together

Use **I clamps** to hold them together



2.

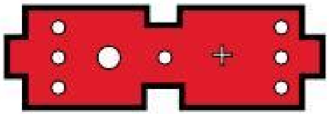
Add the other square as shown

materials needed

wheel - x3

long screw - x1

nut - x1



M-M beam - x1

step 3 : add the counter weight & support

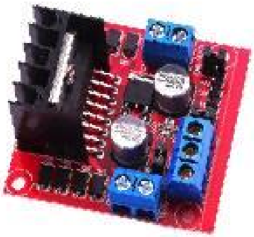
1.

fix three wheels using a long screw to the base, the counter weight will help balance the crane while lifting weight.

2.

add a M-M beam to the base without any clamps or screws

materials needed



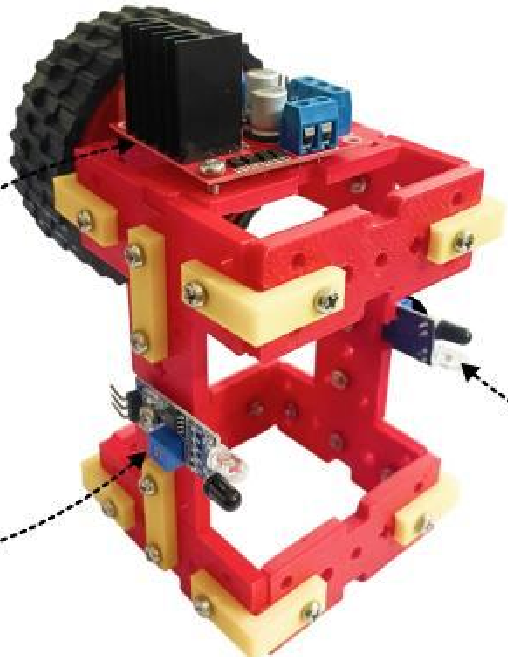
motor driver x1

IR sensor x1

small screw - x3

nut - x3

step 4 : add the motor driver and sensor



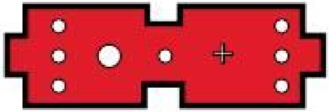
1.

Add the motor driver to the M-M beam, gently screw the motor driver to avoid damaging it

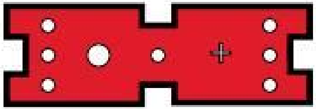
Add one IR sensor here

Add second IR sensor here

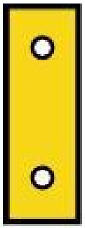
materials needed



M-M beam - x1



F-F beam - x1



I clamp - x2

small screw - x4

nut - x4

step 5 : assemble the servo support

1.

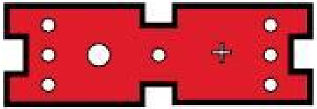
Form a T shape using F-F beam and M-M beam

Use I clamps to hold them together

2.

Fix the T shape to the base as shown

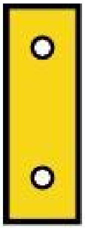
materials needed



F-F beam - x2

servo motor x1

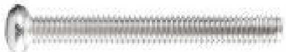
servo arm x1



I clamp - x2

servo screws - x1

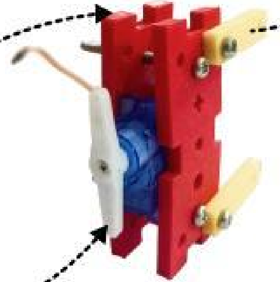
small screw - x4



long screw - x2

nut - x6

step 6 : add the servo motor



1.

Use two F-F beams and fix a servo motor in between using a long screw

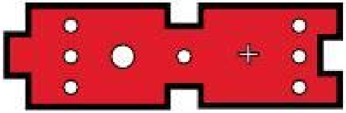
Use a small servo screw to fix the servo arm to the servo motor

2. Add two I clamps to the FF beam

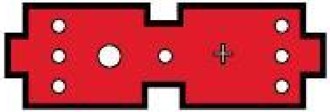
3.

Fix the servo motor to the support as shown

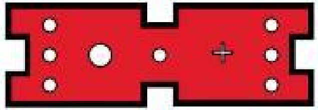
materials needed



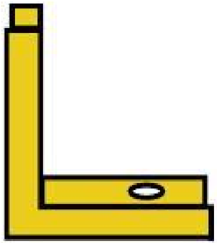
M-F beam - x6



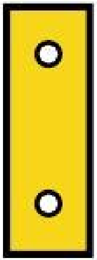
M-M beam - x2



F-F beam - x1



L clamp - x6



I clamp - x1

small screw - x18

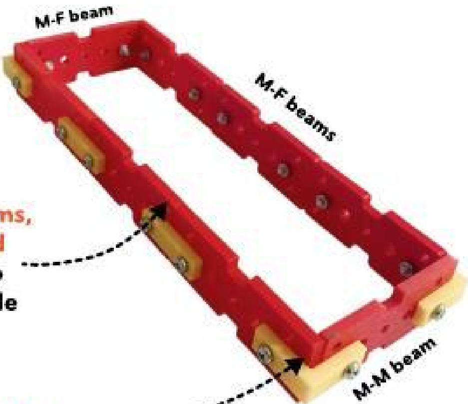
nut - x18

step 7 : assemble the crane‘s jib

1.

Use 6 M-F beams, 1 F-F beam and 1 M-M beam to form a rectangle

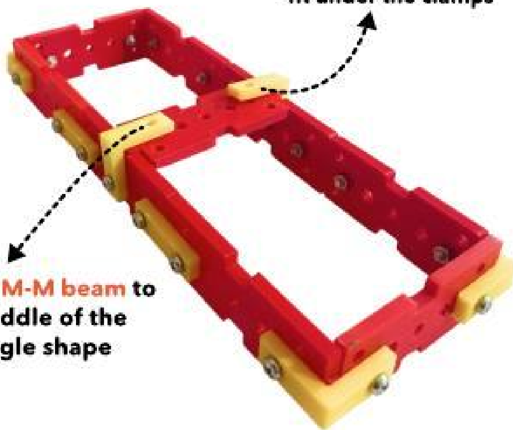
Use I and L clamps to hold it together



Use two L clamps and screw only one hole to allow the servo arm to fit under the clamps

2.

Add a M-M beam to the middle of the rectangle shape



materials needed

motor x1

long screw - x1

servo screws - x2

nut - x1

step 8 : attach the jib and motor

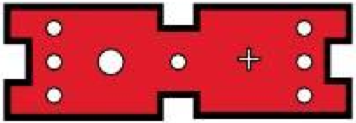
1.

Use servo screw to fix the jib to the arm of the servo motor

2.

Attach the motor to the jib using one long screw

materials needed



F-F beam - x1

coupler - x1

thread x1

step 9 : attach the motor coupler

1.

Fix the motor coupler to the axle of the motor

2.

Tie a thread to the coupler and pass the other end of the thread through the beam

3.

Fix a beam as weight

materials needed

battery x1

battery holder x1

servo controller x1

jumper cables

step 10 : circuit assemble and testing